

LEGEND

- C1

USB 3.0 Standard-B edge X-, centre 39.01, body 13.00 wide, slot -0.80 to 12.60 about the PCB top face
- C2

5 V DC barrel jack edge X-, centre 58.24, body 9.40 wide, slot -0.60 to 11.40 about the PCB top face
- C3

SMA, TX edge X+, centre 20.45, body 9.50 wide, bored 11.00 dia, axis 0.45 above the PCB top face, slotted vertically
- C4

SMA, RX edge X+, centre 60.33, body 9.50 wide, bored 11.00 dia, axis 0.45 above the PCB top face, slotted vertically
- C5

SMB reference clock, right angle edge Y+, centre 91.25, body 8.00 wide, bored 10.00 dia, axis 5.00 above the PCB top face, slotted vertically
- U1

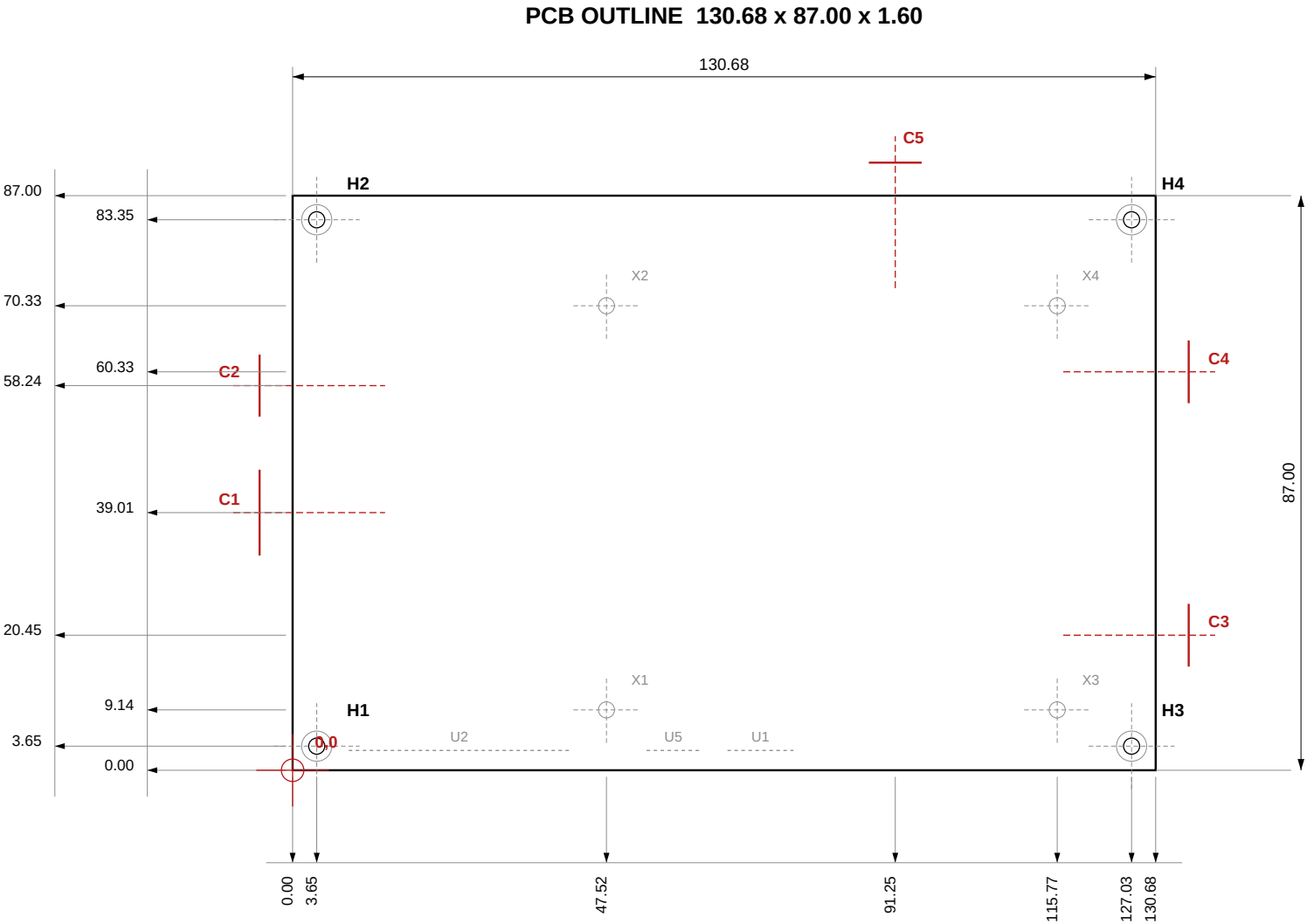
Third SMA position, unfitted edge Y-, centre 70.84
- U2

Expansion header, 2 x 20 at 1.27 mm edge Y-, centre 25.25
- U3

U.FL alternate, 2.1 mm inboard edge X+, centre 34.79
- U4

U.FL alternate, 2.1 mm inboard edge X+, centre 45.22
- U5

U.FL alternate, 1.4 mm inboard edge Y-, centre 57.66



H1 to H4 mounting holes used by this case. 2.44 drill shown solid,
0.180 in (4.57) DXF pad shown grey. Nuand specify 2-56 screws,
so the drill is a #2 clearance. Measure yours to confirm.

X1 to X4 further DXF holes, most likely XB-200 standoffs. Not used.

Red connector body centre and width at the board edge.

Grey features that get no wall opening.

PCB FEATURES (datum = bottom-left PCB corner)

Ref	Feature	X	Y	Note
H1	Mounting hole, case screw	3.65	3.65	2.44 clr
H2	Mounting hole, case screw	3.65	83.35	2.44 clr
H3	Mounting hole, case screw	127.03	3.65	2.44 clr
H4	Mounting hole, case screw	127.03	83.35	2.44 clr
X1	Mounting hole, unused	47.52	9.14	XB-200
X2	Mounting hole, unused	47.52	70.33	XB-200
X3	Mounting hole, unused	115.77	9.14	XB-200
X4	Mounting hole, unused	115.77	70.33	XB-200
C1	USB 3.0 Standard-B	0.00	39.01	body 13.00
C2	5 V DC barrel jack	0.00	58.24	body 9.40
C3	SMA, TX	130.68	20.45	body 9.50
C4	SMA, RX	130.68	60.33	body 9.50
C5	SMB reference clock, right angle	91.25	87.00	body 8.00
U1	Third SMA position, unfitted	70.84	0.00	no opening
U2	Expansion header, 2 x 20 at 1.27 mm	25.25	0.00	no opening
U3	U.FL alternate, 2.1 mm inboard	130.68	34.79	no opening
U4	U.FL alternate, 2.1 mm inboard	130.68	45.22	no opening
U5	U.FL alternate, 1.4 mm inboard	57.66	0.00	no opening

CASE SUMMARY

Overall 138.28 x 94.60 x 25.60
Cavity 132.28 x 88.60, wall 3.00, corner R3.00
Outer edge chamfers bottom 0.80, top 1.20, both 45 deg
Fasteners 4 x M2 x 20 through lid into M2 inserts, bore 3.20 x 4.00 deep
Head recess 4.40 dia x 2.80 deep, seat z 22.80, 3.60 of thread in the insert

WALL OPENINGS (datum = outside corner of the case)

Ref	Edge	Form	Centre	Size	Height range
C1	X-	rect	42.81	14.60 wide	7.20 to 20.60
C2	X-	rect	62.04	11.00 wide	7.40 to 19.40
C3	X+	round	24.25	11.00 dia + 2.55 slot	2.95 to 16.50
C4	X+	round	64.13	11.00 dia + 2.55 slot	2.95 to 16.50
C5	Y+	round	95.05	10.00 dia + 2.00 slot	6.00 to 18.00

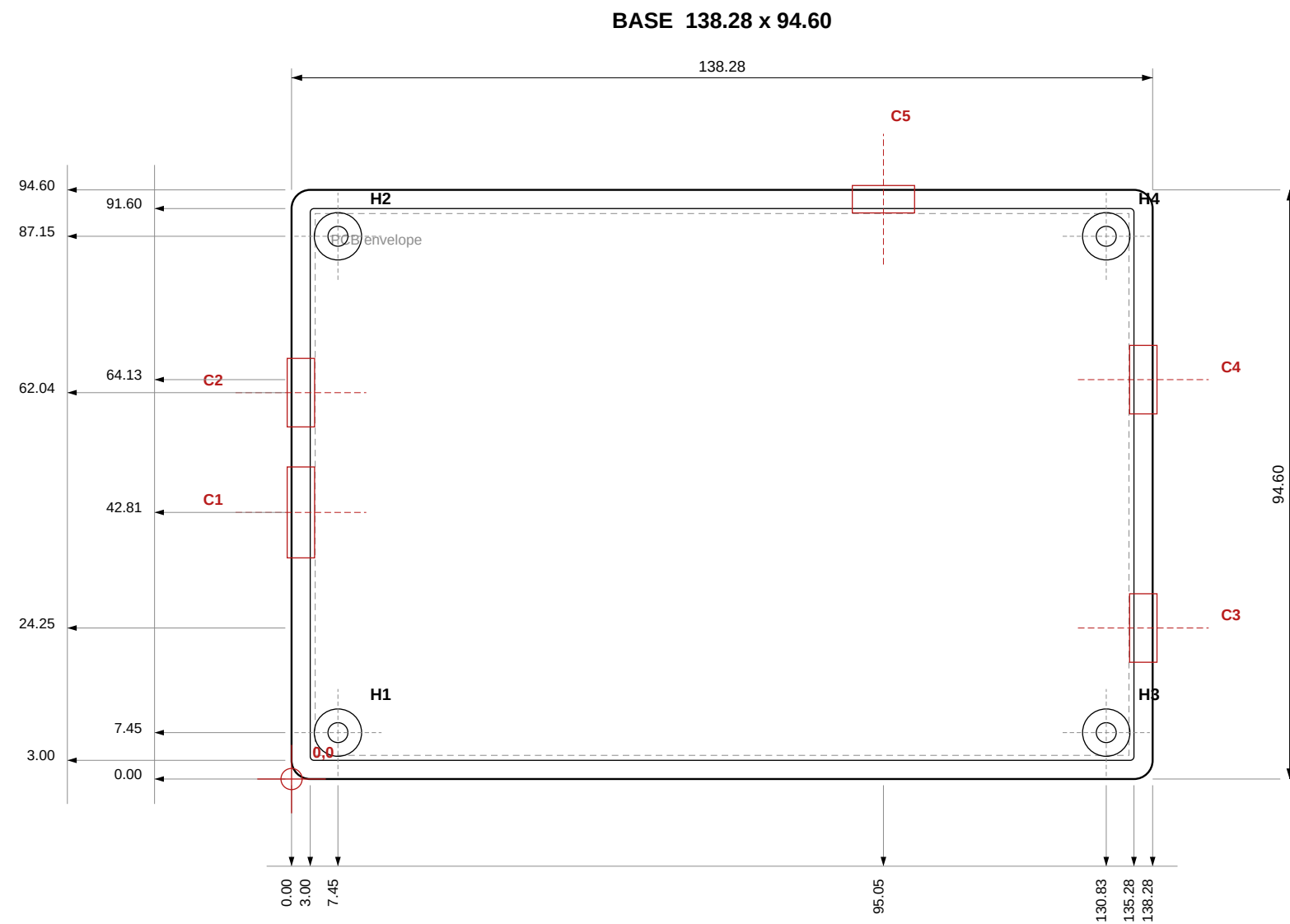
Rectangular openings also get an outer relief pocket 2.00 wider on every side, 2.00 deep, for cable overmoulds. Bored holes get none.
C3 and C4 are bored on the SMA block centre, 0.45 above the PCB top face: a 9.50 x 8.10 block, 4.50 up and 2.00 down off the board.
Bored openings are drawn out into a vertical slot on the same diameter: the round end stays on the connector axis and the far end sits on the parting plane at 11.00, so no half carries the arc's level tangent, which would print unsupported on a wall that stands vertical on the bed. The height range column is the full slot.

HEIGHT STACK-UP (from outside of base floor)

Outside of floor	0.00
Inside of floor	2.40
Insert bore floor	2.40
Screw tip, fully driven	2.80
Standoff top, PCB underside	6.40
PCB top surface	8.00
Parting plane, base to lid	11.00
Head seat, recess floor	22.80
Cavity ceiling	23.00
Outside of lid	25.60

OPENING REFS

- | | | |
|----|----------------------------------|--|
| C1 | USB 3.0 Standard-B | centre 42.81, 14.60 wide, z 7.20 to 20.60 |
| C2 | 5 V DC barrel jack | centre 62.04, 11.00 wide, z 7.40 to 19.40 |
| C3 | SMA, TX | centre 24.25, 11.00 dia slot, axis z 8.45, ends z 8.45 and 11.00 |
| C4 | SMA, RX | centre 64.13, 11.00 dia slot, axis z 8.45, ends z 8.45 and 11.00 |
| C5 | SMB reference clock, right angle | centre 95.05, 10.00 dia slot, axis z 13.00, ends z 11.00 and 13.00 |

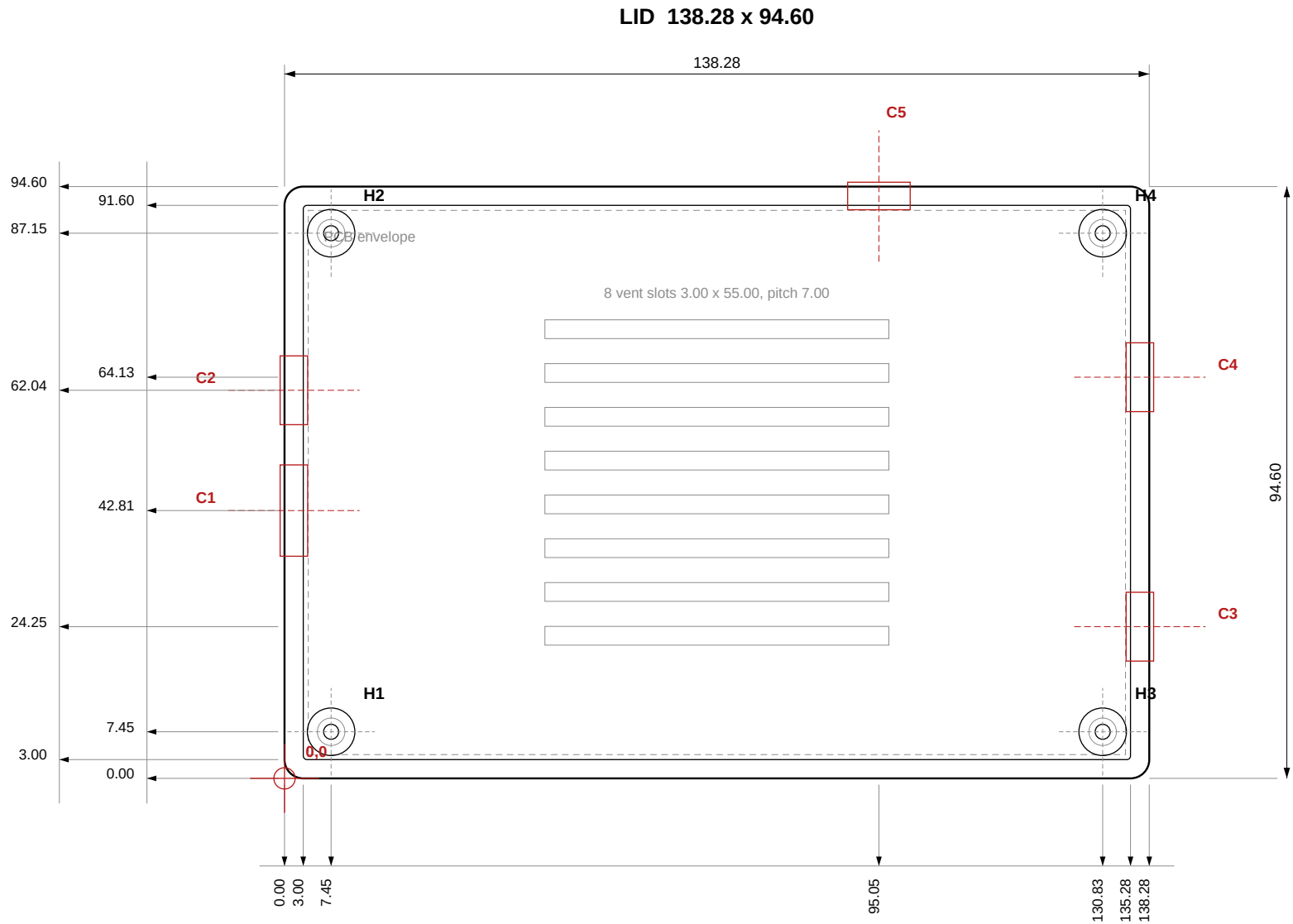


NOTES

Standoff OD 7.60, height 4.00 above the floor.
Insert bore 3.20 dia x 4.00 deep for M2 heat-set inserts.
Wall reduced to 1.25 for the top 3.00 to form the lip rebate.
Base height 11.00. Floor 2.40.
Bottom outer edge chamfered 0.80 x 45 deg. See sheet 5.

OPENING REFS

- C1 USB 3.0 Standard-B centre 42.81, 14.60 wide, z 7.20 to 20.60
- C2 5 V DC barrel jack centre 62.04, 11.00 wide, z 7.40 to 19.40
- C3 SMA, TX centre 24.25, 11.00 dia slot, axis z 8.45, ends z 8.45 and 11.00
- C4 SMA, RX centre 64.13, 11.00 dia slot, axis z 8.45, ends z 8.45 and 11.00
- C5 SMB reference clock, right angle centre 95.05, 10.00 dia slot, axis z 13.00, ends z 11.00 and 13.00



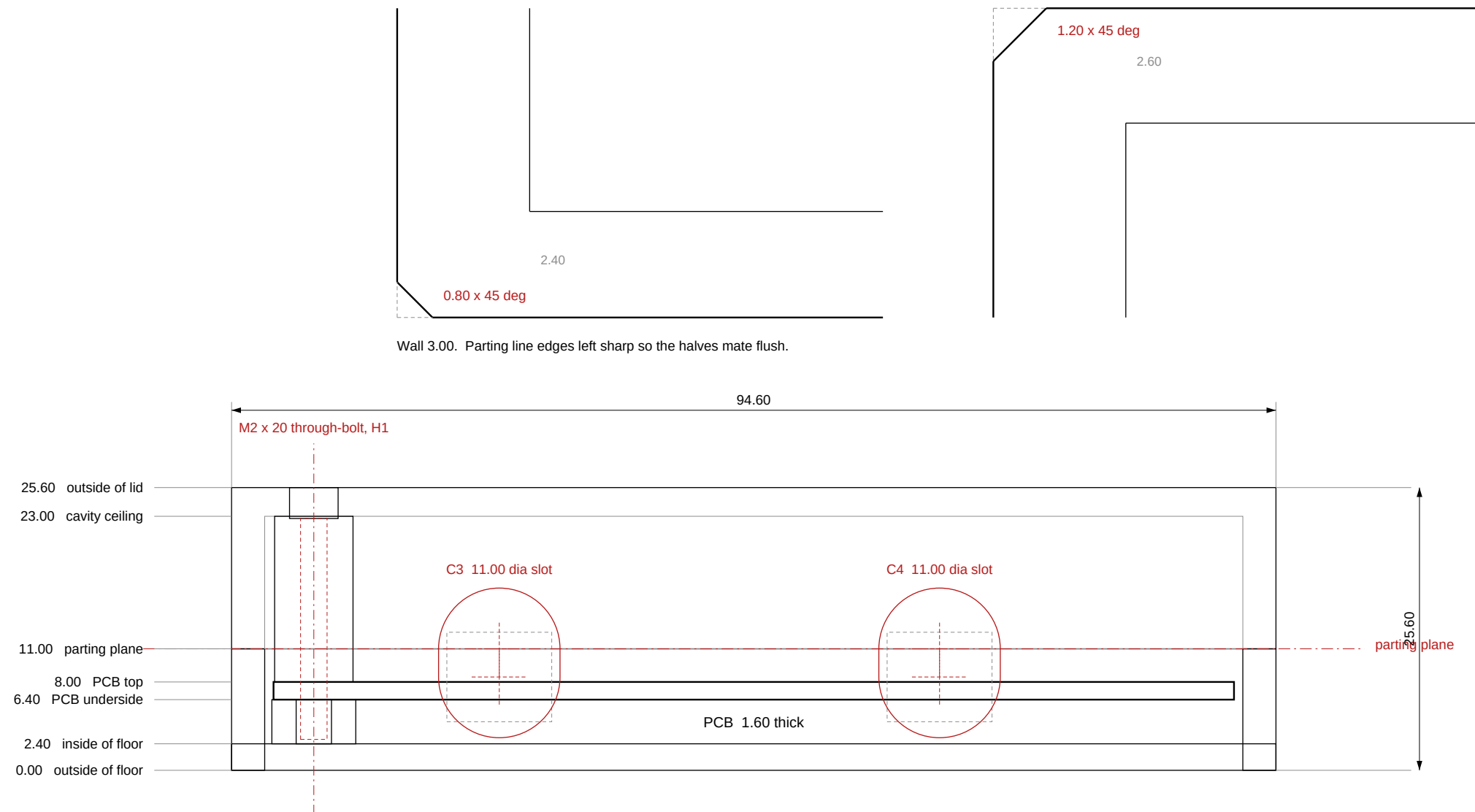
NOTES

- Boss OD 7.60, from PCB top face to cavity ceiling.
- Screw 2.40 clr, head recess 4.40 dia x 2.80 deep, seat z 22.80, 45 deg cone under.
- Recess depth follows the M2 x 20 screw, not the plate thickness.
- Tip stops 0.40 above the insert bore floor, 3.60 of thread engaged.
- Lip 1.20 thick x 3.00 deep, 0.20 slip fit, 0.35 relief off the cavity wall.
- Lid height 14.60 above the parting plane. Top plate 2.60.
- Top outer edge chamfered 1.20 x 45 deg. See sheet 5.

DETAIL OUTER EDGE CHAMFERS 8:1

Base, bottom outer edge

Lid, top outer edge



Cut plane runs parallel to the short PCB edge, 7.45 from the case datum, through openings C3 and C4 and the H1 screw column.

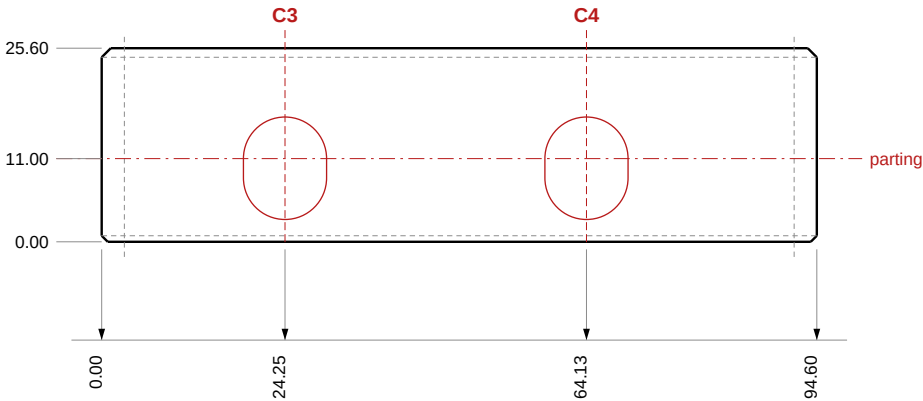
Drawn at 2:1. Values on the left are heights above the outside of the base floor.

M2 x 20 screw: head seat z 22.80, recess 2.80 deep, tip z 2.80, 3.60 of thread inside the 4.00 deep insert bore.

C3 and C4 bored 11.00 dia about the SMA block centre, 0.45 above the PCB top face, then drawn up to the parting plane as a slot on the same diameter. Grey dashed outline is the 9.50 x 8.10 block, 4.50 above and 2.00 below the board. Crosses mark the two slot end centres.

WALL X+ short edge, SMA end

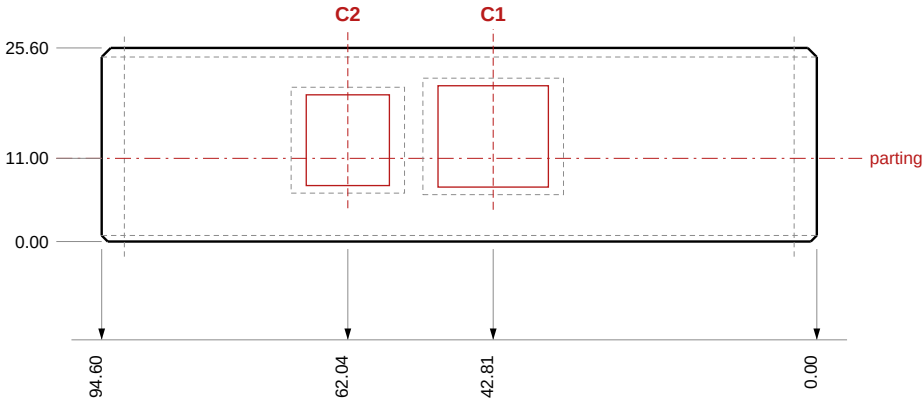
Outside face at 1:1, +y to the right



- C3** SMA, TX
centre y 24.25, 11.00 dia slot, 2.55 straight
opening z 2.95 to 16.50, spans the parting plane
- C4** SMA, RX
centre y 64.13, 11.00 dia slot, 2.55 straight
opening z 2.95 to 16.50, spans the parting plane

WALL X- short edge, USB and power end

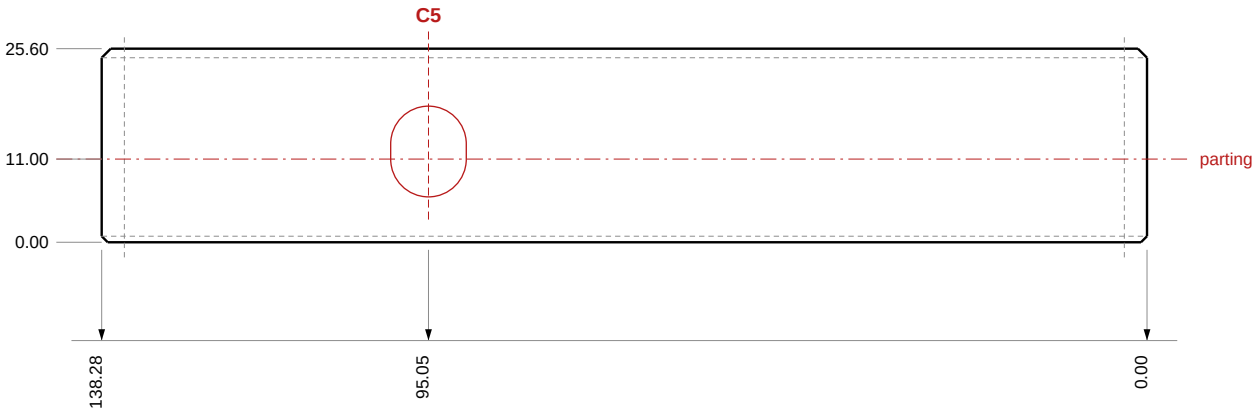
Outside face at 1:1, +y to the left



- C1** USB 3.0 Standard-B
centre y 42.81, 14.60 wide, plus 2.00 relief all round 2.00 deep
opening z 7.20 to 20.60, spans the parting plane
- C2** 5 V DC barrel jack
centre y 62.04, 11.00 wide, plus 2.00 relief all round 2.00 deep
opening z 7.40 to 19.40, spans the parting plane

WALL Y+ long edge, reference clock

Outside face at 1:1, +x to the left



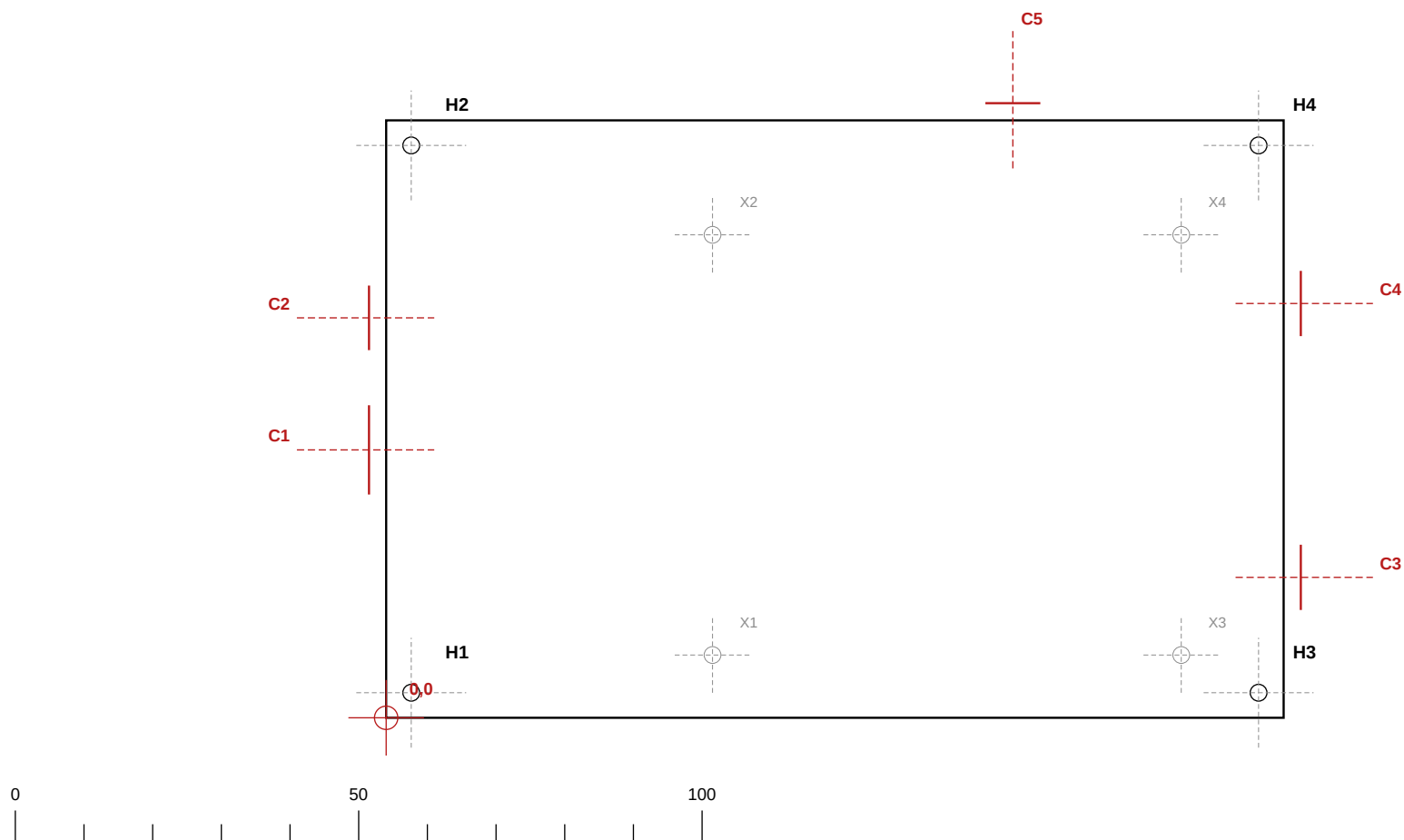
- C5** SMB reference clock, right angle
centre x 95.05, 10.00 dia slot, 2.00 straight
opening z 6.00 to 18.00, spans the parting plane

NOTES

Each wall is drawn as seen from outside the case, at 1:1, so the printed part can be held against the page. Two of the three views therefore run their datum axis right to left; the figures under every view are absolute datum coordinates either way.

Grey dashed verticals are the corner radius tangents - every opening must stay between them, on the flat. Grey dashed horizontals are the 0.80 bottom and 1.20 top chamfer tangents. Y- carries no opening; the expansion header stays inside the case.

Material below the red parting line at 11.00 is base, above it is lid.



SCALE CHECK. This bar must measure exactly 100.0 mm.

If it does not, reprint with scaling set to 100 percent or actual size.

Lay the PCB on the outline. Every hole centre and connector centreline should sit under its mark.

Ticks outside the outline show connector body width; the dashed line is its centre.